Introduction

> *I can't teach you mathematics with this book, but I believe that much hard work on your part, chipping away at the problems, will eventually reveal the mathematician that is within you.*
>
> — Stephen Siklos, *Advanced Problems in Mathematics*

This is a book of hard problems, and of what lies behind them. Every one of its 136 sections starts with a question from a past STEP paper. But the question is only the way in. Behind almost every good STEP question is a piece of mathematics that you will meet properly at university, and the aim of this book is to take you through the door and show you what is on the other side.

What STEP is, and why it exists

STEP, the Sixth Term Examination Paper, is a set of mathematics examinations taken in the summer of the final year of school. It was introduced by the University of Cambridge in 1987, and its job is to help decide who should study mathematics there: almost all Cambridge offers for Mathematics are conditional on STEP grades, and over the years other universities, notably Warwick, have used it too. At the start there were STEP papers in many subjects, but by 2001 only mathematics remained. For most of its history there were three papers, I, II and III; Paper I was last set in 2019, and today candidates take Papers II and III. The examination was run for many years by Cambridge Assessment, and since 2024 it has been administered by OCR.

A STEP paper looks nothing like a school examination. Instead of ten short, compulsory questions, there are a dozen or so long ones, of which candidates attempt six, and a strong candidate who makes real progress on four will do very well. The questions do not tell you how many marks each part is worth, and they rarely ask you to do something you have practised. They ask you to take the mathematics you already know and use it somewhere new.

Stephen Siklos, who knew STEP better than anyone, described its purpose like this. It is a hurdle: success in STEP turns out to predict success in a university mathematics degree remarkably well (he reported a correlation of 0.63 between STEP rank and first-year Cambridge examination rank, very high by the standards of such things). It is a preparation: the style of thinking that STEP rewards is the style of undergraduate mathematics. And it is a test of motivation: preparing for STEP takes real effort, and a student who is not willing to make that effort is unlikely to enjoy a demanding mathematics degree.

Stephen Siklos

It is impossible to write about STEP without writing about Stephen Siklos. He was a Cambridge mathematician, a former research student of Stephen Hawking who worked on the mathematics of the early universe and of gravitational waves, a lecturer in the Department of Applied Mathematics and Theoretical Physics, and later Senior Tutor and then President of Jesus College. He became the Senior Examiner for STEP Mathematics when it began in 1987, and held the post for thirty-two years, until his death in August 2019. In that time he drafted questions, "knocked other people's questions into shape" (his own phrase, for something he said had long been a source of interest and entertainment), scrutinised every paper and oversaw the marking. He was also the driving force behind the STEP Easter School and the STEP Support Programme, which give free help to students whose schools cannot. One obituary said that he did more than anyone to make the Cambridge entrance examination in mathematics accessible to students whatever their school background.

His book *Advanced Problems in Mathematics: Preparing for University*, published free online by Open Book Publishers, is still the best introduction to STEP, and I strongly recommend it. Two things he wrote in it have shaped this book. The first is his view of what a good STEP question is: one in which the first part shows you an unfamiliar idea, and the later parts ask you to show that you have understood it by using it again in a harder setting; or one whose answer depends on a parameter that you have to discover for yourself; or one that starts with special cases and ends with a general result. Those are exactly the questions that lead somewhere, and most of the questions in this book are of those kinds. The second is his insistence that solutions should be "springboards rather than feather beds". You learn nothing by reading a solution you have not first fought with yourself.

How I met STEP

I first came across STEP in the early 1990s, when my mathematics teacher gave me some of the papers as an extra challenge. Thankfully, I never had to sit the real examination. I remember those papers as being extraordinarily hard: questions that seemed to have nothing to do with anything I had learned, that went on for a page, and that I could start but rarely finish. I also remember the feeling, on the occasions when something finally worked, that I had seen something that school mathematics had not shown me.

Many years later I came back to STEP from the other side, helping students to prepare for it. Working through hundreds of questions with them, I came to see the papers differently. Not every question is a gem. Some are, frankly, long tests of stamina: a page of careful algebra, a demanding but routine calculation, with not much insight or pleasure at the end. Stamina matters, and university mathematics needs it, but those questions are not what make STEP special.

What makes it special is that, scattered across the many thousands of questions set since 1987, there are hundreds of genuinely beautiful ones. The best way to explain what I mean is to show you some.

Some gems

Here are seven of the questions you will meet later in the book, chosen to span its range: two from combinatorics, two from dynamical systems, two from mechanics and one from probability. Each starts with something you can picture, ends with something you might not have believed, and opens onto a piece of mathematics that is studied seriously at university.

Order hiding in chaos. Draw six dots on a page and join every pair with a line, fifteen lines in all, colouring each line gold or silver however you like. Now try to avoid creating a triangle whose three sides are all the same colour. You will fail, every time. With five dots it can be done (colour the sides of a pentagon gold and the five diagonals silver), but with six it is impossible, however cunning your colouring. A 1989 STEP I question asks you to prove this, with a single hint: look at the five lines that leave one dot. Three of them must share a colour, and from there the proof takes a few lines. There are $2^{15} = 32\,768$ ways to colour the lines, and the argument deals with all of them at once, without checking a single case.

The result is the smallest case of Ramsey's theorem, proved by Frank Ramsey in 1928, which says that complete disorder is impossible: any large enough structure, however haphazardly it is put together, contains a pocket of perfect order. How large is "large enough"? To guarantee four dots joined entirely in one colour you need 18 dots, a fact proved in 1955. For five dots, nobody knows: the answer lies somewhere between 43 and 46, and the upper bound was proved only in 2024. Paul Erdős joked that if aliens threatened to destroy the Earth unless we told them this number, we should set every computer and mathematician to work on it; but if they asked for the answer for six, we should try to destroy the aliens. You will meet Ramsey theory in a university combinatorics or graph theory course, and the pigeonhole principle that powers the STEP proof in almost every course you take (Section 9.2).

Strange dice. With two ordinary dice, a total of 7 is the most likely and totals of 2 and 12 the least, in the familiar pattern of $1, 2, 3, 4, 5, 6, 5, 4, 3, 2, 1$ ways. Is there any other way to number a pair of dice with positive whole numbers that gives exactly the same chances for every total? It seems unlikely, but there is exactly one: $1, 2, 2, 3, 3, 4$ on one die and $1, 3, 4, 5, 6, 8$ on the other. These are *Sicherman dice*, named after George Sicherman, who found them, and made famous by Martin Gardner in 1978. In any game that depends only on the total thrown, you could swap them for ordinary dice and nobody would ever know.

A 1993 question approaches the idea from the other end. It gives you a table showing how many ways each total can be thrown with two oddly numbered dice, and asks you to work out the numbering, like a detective reconstructing a crime from its traces. The section then shows the idea that makes the whole subject simple: turn each die into a polynomial, with a term x^k for each face showing k. Throwing two dice corresponds to multiplying their polynomials, because $x^i \times x^j = x^{i+j}$ adds the numbers on the faces. Finding new dice with the same totals becomes a matter of factorising a polynomial in a different way, rather as 36 can be split as 4×9 instead of 6×6. And because polynomials, like whole numbers, factorise in only one way into pieces that cannot be

broken down further, the factorisation also proves that Sicherman's dice are the only alternative. This trick of encoding a counting problem in a polynomial, a *generating function*, is one of the most powerful tools in combinatorics, and the unique factorisation behind it is a central theme of a first university course in algebra (Section 9.6).

Chaos with a formula. Start with a number between 0 and 1, multiply it by 4 and by one minus itself, and repeat: $u_{n+1} = 4u_n(1 - u_n)$. This is the *logistic map*, a crude model of a population that grows but is limited by its food supply. The rule could hardly be simpler, yet the numbers it produces jump around with no visible pattern, and two starting values that agree to ten decimal places soon give completely different sequences. The biologist Robert May drew attention to it in 1976, in a famous paper with the title "Simple mathematical models with very complicated dynamics", and it became the standard example of chaos, usually explored by computer.

A 2014 question reveals something unexpected: the sequence has an exact formula. If the starting value is $\sin^2\theta$, then $u_n = \sin^2(2^n\theta)$, which you can prove with nothing more than the double angle formulae and induction. (The formula was known to Stanisław Ulam and John von Neumann in the 1940s.) You might expect an exact formula to tame the chaos. Instead it explains it completely. Each step doubles the angle, so any tiny uncertainty about where you started doubles at every step; after fifty steps it has grown about a million billion times. Writing the angle, as a fraction of π, in binary makes this vivid: each step of the map shifts the binary digits one place to the left and throws away the first, so the sequence reads off the digits of the starting value one by one, and to predict a hundred steps ahead you need to know the starting value to a hundred binary places. The same formula also finds every cycle of the map, and shows that there are cycles of every length, packed densely into $[0, 1]$, every one of them unstable. This is the subject of dynamical systems, which you will meet in university courses on differential equations and, later, in courses on chaos itself (Section 8.7).

Period three implies chaos. Take the function $f(x) = 7 - 2|x|$, start with a number, and keep feeding the answer back in. Starting at 1 gives $1, 5, -3, 1, 5, -3, \ldots$: a cycle of length three. A 2025 question asks you to hunt for cycles like this one, by sketching the graphs of f, $f(f(x))$ and $f(f(f(x)))$ and finding where they cross the line $y = x$. Each extra application of f doubles the number of zigzags in the graph, and the section turns that observation into a proof that this function has cycles of *every* length: two, three, four, a hundred, a million.

That is not a special feature of this function. In 1975 Tien-Yien Li and James Yorke proved that *any* continuous function with a cycle of length three must have cycles of every other length, in a paper with the irresistible title "Period three implies chaos". The proof needs nothing more than the intermediate value theorem, which you already use whenever you argue that a continuous function that changes sign must cross zero, together with a clever diagram showing which intervals the function stretches over which. It later turned out that a Ukrainian mathematician, Oleksandr Sharkovsky, had proved a stronger result in 1964: he found a strange ordering of the whole numbers, beginning $3, 5, 7, 9, \ldots$, such that a cycle of any length forces cycles of every length that comes after it. The intermediate value theorem is one of the first results proved carefully in a university analysis course; this question shows how much it can do (Section 8.8).

Why a bicycle pump gets warm. Pump up a bicycle tyre and then feel the pump: it is warm. Squeezing a gas heats it, which is how a diesel engine ignites its fuel without a spark. But why? A 2002 question answers this by following a single air molecule as it bounces back and forth between the end of the pump and the piston being pushed in. Each time the molecule bounces off the advancing piston, it picks up a little speed, like a tennis ball hit by a moving racket. The question asks you to describe one bounce exactly, turn that into a recurrence for the length of the cylinder at each collision, and then solve the recurrence.

The solution reveals a hidden law: the molecule's speed multiplied by the length of the cylinder hardly changes. Halve the length and you double the speed, and since temperature measures the energy of the molecules, the gas heats up. Quantities like this, which stay almost constant when a system is changed slowly, are called *adiabatic invariants*, and with a little more work the same argument leads to the law of adiabatic compression of a gas that you will meet in university thermodynamics. The idea reaches a long way. Enrico Fermi suggested in 1949 that cosmic rays reach their enormous energies in exactly this way, by bouncing repeatedly between moving magnetic fields in space; and Paul Ehrenfest argued in 1916 that adiabatic invariants are the quantities that quantum mechanics must quantise. A question about a bicycle pump turns out to touch statistical physics, astrophysics and the beginnings of quantum theory (Section 10.4).

A ball that comes back. Throw a ball gently forwards and it moves steadily away from you until it lands. Common sense says this is always so: the ball is always moving forwards, so how could it get closer? But throw it steeply enough, at more than about 70.5° to the horizontal, and something odd happens. For part of its flight, as it comes down, the ball gets *closer* to you, even though it is still moving away horizontally. Its height is falling faster than its horizontal distance is growing.

A 2019 question asks you to prove this and to find the critical angle exactly. The method is one you will use again and again: instead of the distance itself, differentiate its square, which avoids square roots; the derivative turns out to be a quadratic in time, and the question of whether the distance ever decreases becomes the question of whether that quadratic has real roots. Its discriminant gives the answer, and the critical angle turns out to be the angle whose sine is exactly $\frac{2\sqrt{2}}{3}$. The lesson goes well beyond projectiles. Many quantities that "obviously" increase do not, and at university, showing that something is monotonic, by finding the sign of a derivative, is a key step in proving inequalities, in proving that sequences converge, and in showing that the solutions of a differential equation settle down (Section 11.10).

The last passenger. A train is fully booked. The first passenger to board ignores their ticket and sits in a seat chosen at random. Everyone after that sits in their own seat if it is free, and in a randomly chosen free seat if it is not. What is the chance that the last passenger to board finds their own seat free? With a hundred seats the possibilities branch bewilderingly: one displaced passenger can displace another, who displaces another, in a chain of any length. A direct calculation looks hopeless. Yet the answer is exactly $\frac{1}{2}$, whatever the number of seats.

A 2021 question reaches this answer step by step: it asks you to find the answer for two

and three seats, to set up a recurrence by conditioning on where the first passenger sits, and then to prove by induction that the answer is always $\frac{1}{2}$. That is a perfectly good proof, but it leaves you wondering *why* the answer is so simple, and the section then gives a one-line argument that explains it. Whenever a passenger has to choose at random, the first passenger's seat and the last passenger's seat are treated in exactly the same way; the process cannot tell them apart until one of them is taken. So the last free seat is equally likely to be either, and it must be one of them, because every other seat is taken by its owner, or by someone displaced, before the last passenger boards. Arguments from symmetry like this, which replace pages of calculation with a single observation, run all through university probability (Section 12.5).

Taken together, questions like these touch every one of the main areas of a first-year mathematics degree: number theory, analysis, calculus, complex numbers, algebra, linear algebra, differential equations, combinatorics, mechanics, probability and statistics.

What this book is for

That observation is the reason for this book. Most STEP preparation, quite reasonably, aims at passing the examination: learn the techniques, practise the papers, build up speed and stamina. This book aims at that too, and working through it will make you better at STEP. But its real purpose is different. It is to help you prepare for STEP *with university mathematics in mind*: to see each question not as an obstacle to be cleared but as an invitation, and to follow the invitation as far as it goes.

The students who do best at STEP, and who flourish at university afterwards, are almost always the ones who love mathematics for its own sake. They enjoy the moment when a hard problem gives way. They want to know why a trick works, not just that it does. When an answer comes out to be suspiciously simple, like $\frac{1}{2}$ or $\frac{1}{e}$ or $\pi^2/6$, they suspect that something deeper is going on, and they go looking for it. Those are the habits this book tries to encourage. Each question is chosen because it leads to a discovery: an idea, a theorem, a way of thinking, that is bigger than the question itself.

The ways of thinking matter as much as the theorems. University mathematics asks you to think in ways that school mathematics rarely does: to prove things rather than calculate them, to argue by contradiction or by infinite descent, to count the same thing in two ways, to find a quantity that never changes, to construct a counterexample, to show that something exists without finding it. STEP questions train exactly these habits, often without saying so. In this book every section names the thinking skill at its heart, so that you can recognise it when you meet it again.

How the book is organised

The book has thirteen chapters, organised by the area of university mathematics that the questions lead to, rather than by the STEP paper they came from.

- **Pure mathematics** (Chapters 1 to 9): number theory and proof; sequences, series and convergence; functions, inequalities and rigour; calculus done properly; complex numbers; algebra and structure; linear algebra and geometry; differential equa-

tions and dynamical systems; and counting and chance, which also draws on the combinatorial questions from the probability papers.

- **Mechanics** (Chapters 10 and 11): conservation laws, collisions and gravity; and oscillations, rotation and moving frames, ending with a set of surprising trajectories.
- **Probability and statistics** (Chapters 12 and 13): probability, from clear thinking about conditioning to the central limit theorem; and statistics and randomness in the world.

Each chapter opens with an essay: why the area matters, what you will meet in it at university, and three big questions that run through the chapter. Each closes with a wrap-up that returns to those questions, a toolkit of the methods used, pointers to other chapters, and suggestions for reading before university. At the end of each chapter are brief answers to the exercises. Before Chapter 1 there is a glossary of the terms in the book that are unlikely to be familiar from school, and a table of the mathematical symbols it uses; if a word or a symbol stops you, look it up there.

Every section follows the same pattern. It begins with *the question*, exactly as it was set. *The hook* explains why the question is more interesting than it looks. *Before you start* gives you a first foothold. *Working it through* is a guided solution in stages, interrupted by boxes headed *Ask yourself* and *Pause and try*, which are the questions a good teacher would ask you at that point; they are there to be answered, not skipped. *Looking back* picks out the key moves. *What's really going on* is the heart of the section: the university mathematics behind the question, its history and its surprises. *The thinking skill* names the habit of mind the question exercises. *Try this next* gives follow-up problems, often including related STEP questions, and *Further reading* suggests books and articles at a level a strong school student or first-year undergraduate can manage.

A few sections use topics that are not on the current STEP specification, such as groups, moments of inertia and moment generating functions. They are included because the questions are too good to leave out, and each gives the facts it needs.

How to use this book as a student

- **Do the question first.** Give each question at least half an hour of honest effort before you read further. Being stuck is not a sign of failure; it is where the learning happens. If you are completely stuck, read *Before you start*, then try again.
- **Answer the boxes.** When the text asks you a question, stop and answer it, on paper. The guided solutions are written to be read with a pencil in your hand.
- **Do not skip the "What's really going on" sections.** They are not examinable, but they are the reason the book exists, and they are what you will remember.
- **Read the sections in any order.** Each section stands on its own, and cross-references point you to related ideas elsewhere. Follow your curiosity.
- **Try the exercises, then check the answers.** The answers at the end of each chapter are brief: they give the result and the key step, not a full write-up.

- **Use the worksheets.** The website has a printable worksheet for every section, with the question, the hints in order and the exercises, for working away from a screen.
- **Read beyond the book.** Siklos's book, the STEP Support Programme and the past papers themselves are all freely available. The further reading in each section is chosen to be accessible before university.

How to use this book as a teacher

- **As a source of extension work.** Each section is a self-contained session: the question, a guided path through it, and somewhere interesting to go afterwards. The thinking skills make good themes for a sequence of sessions.
- **As a discussion guide.** The *Ask yourself* and *Pause and try* boxes are natural prompts for a class or small group. Pose them before revealing the next stage.
- **With the worksheets.** Each worksheet gives the question, the hints in order and the follow-up exercises; students can choose how many hints to take.
- **For STEP preparation.** Questions from all three papers, from the specimen papers to 2025, are included, many with related questions in the exercises. The chapter wrap-ups and toolkits help students to see connections across topics.
- **For students without STEP support.** The book is written so that a determined student can work through it alone. If your school does not offer STEP preparation, it can do much of that job.

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I have tried hard to make the mathematics correct, but a book of this size will certainly contain errors. If you find one, however small, please report it as an issue on the book's GitHub repository, `github.com/keithmcnulty/step_book_mcnulty`. Corrections will be made on the website, `beyondstep.keithmcnulty.org`, as they come in.

A last word before you start

Siklos ended the introduction to his book with the story of Michelangelo's *David*, carved from a block of marble that had lain untouched for decades, and said that he could not teach you mathematics, but that hard work chipping away at problems would reveal the mathematician within you. I cannot improve on that. This book offers you the marble, and some tools, and a few pictures of what other people have found inside. The chipping is up to you. I hope you enjoy it as much as I have enjoyed putting the book together.

Keith McNulty
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